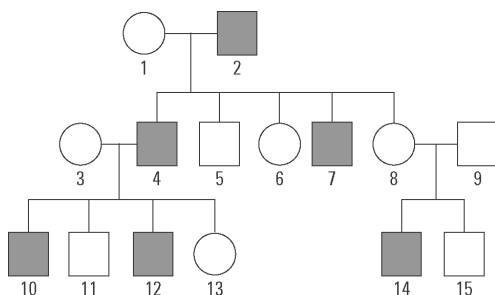


Pedigrees

Student: Class:

1. Consider the following pedigree for three generations of a family. The shaded squares show those individuals who have inherited a particular trait or characteristic. The characteristic is not sex-linked.



- (a) How is person 5 related to person 9?

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- (b) What relation is person 8 to person 13?

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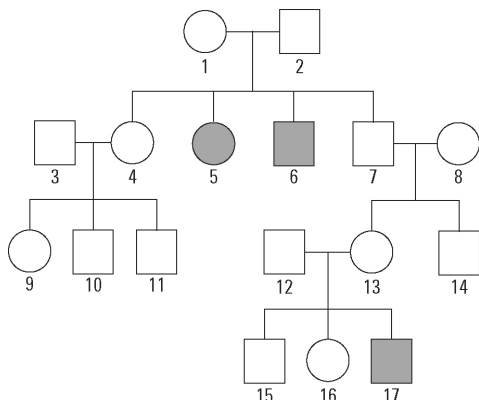
- (c) Is this trait dominant or recessive? Justify your answer.

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- (d) Identify those who are definitely carriers of the gene for this trait but who are not affected.

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2. The following pedigree shows the inheritance of cystic fibrosis. The shaded individuals have the disease.



- (a) Is the gene for cystic fibrosis likely to be dominant or recessive based on this pedigree?

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- (b) Some individuals are carriers of the gene but do not have the disease. Identify which individuals are carriers.

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- (c) If individuals 12 and 13 have more children, what is the chance that any future child will have cystic fibrosis? Justify your answer.

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